Use Cases

for

PathSG

Version 1.4

Prepared by Paul Tamoghna

PathSG

8 Nov 2026

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Revision History

Name	Date	Reason For Changes	Version
Max Lian	05 Sep 2026	Initial Use Case Description	V1.0
Max Lian	12 Sep 2026	Additional Use Case Description	V1.1
Max Lian	26 Oct 2026	Revised “3.1: Generate Education Roadmap” Revised “5.1: Update School Database” Added “5.2: Export School Database”	V1.2
Paul Tamoghna	5 Nov 2026	Aligned use cases with SRS v1.4. Revised 1.1, 1.2, 2.1–2.7, 3.1–3.3, 4.1–4.4, 5.1, 5.2. Added map display (OneMap), login and AI rate limiting, token-based sessions, ranking priority.	V1.3
Paul Tamoghna	8 Nov 2026	Aligned with SRS v1.4: revised 1.1, 1.2, 2.1, 2.2, 2.4, 3.1, 3.3, 4.1, 4.3, 5.1, 5.2; added Use Case Diagram and created the table of contents page	V1.4

Guidance for Use Case Template

Document each use case using the template shown in the Appendix. This section provides a description of each section in the use case template.

1. Use Case Identification

1.1. Use Case ID

Give each use case a unique numeric identifier, in hierarchical form: X.Y. Related use cases can be grouped in the hierarchy. Functional requirements can be traced back to a labeled use case.

1.2. Use Case Name

State a concise, results-oriented name for the use case. These reflect the tasks the user needs to be able to accomplish using the system. Include an action verb and a noun. Some examples:

- View part number information.
- Manually mark hypertext source and establish link to target.
- Place an order for a CD with the updated software version.

1.3. Use Case History

1.3.1 Created By

Supply the name of the person who initially documented this use case.

1.3.2 Date Created

Enter the date on which the use case was initially documented.

1.3.3 Last Updated By

Supply the name of the person who performed the most recent update to the use case description.

1.3.4 Date Last Updated

Enter the date on which the use case was most recently updated.

2. Use Case Definition

2.1. Actor

An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks. Different actors often correspond to different user classes, or roles, identified from the customer community that will use the product. Name the actor(s) that will be performing this use case.

2.2. Description

Provide a brief description of the reason for and outcome of this use case, or a high-level description of the sequence of actions and the outcome of executing the use case.

2.3. Preconditions

List any activities that must take place, or any conditions that must be true, before the use case can be started. Number each precondition. Examples:

1. User's identity has been authenticated.
2. User's computer has sufficient free memory available to launch task.

2.4. Postconditions

Describe the state of the system at the conclusion of the use case execution. Number each postcondition. Examples:

1. Document contains only valid SGML tags.
2. Price of item in database has been updated with new value.

2.5. Priority

Indicate the relative priority of implementing the functionality required to allow this use case to be executed. The priority scheme used must be the same as that used in the software requirements specification.

2.6. Frequency of Use

Estimate the number of times this use case will be performed by the actors per some appropriate unit of time.

2.7. Flow of Events

Provide a detailed description of the user actions and system responses that will take place during execution of the use case under normal, expected conditions. This dialog sequence will ultimately lead to accomplishing the goal stated in the use case name and description. This description may be written as an answer to the hypothetical question, "How do I <accomplish the task stated in the use case name>?" This is best done as a numbered list of actions performed by the actor, alternating with responses provided by the system.

2.8. Alternative Flows

Document other, legitimate usage scenarios that can take place within this use case separately in this section. State the alternative course, and describe any differences in the sequence of steps that take place. Number each alternative course using the Use Case ID as a prefix, followed by "AC" to indicate "Alternative Course". Example: X.Y.AC.1.

2.9. Exceptions

Describe any anticipated error conditions that could occur during execution of the use case, and define how the system is to respond to those conditions. Also, describe how the system is to respond if the use

case execution fails for some unanticipated reason. Number each exception using the Use Case ID as a prefix, followed by “EX” to indicate “Exception”. Example: X.Y.EX.1.

2.10. Includes

List any other use cases that are included (“called”) by this use case. Common functionality that appears in multiple use cases can be split out into a separate use case that is included by the ones that need that common functionality.

2.11. Special Requirements

Identify any additional requirements, such as nonfunctional requirements, for the use case that may need to be addressed during design or implementation. These may include performance requirements or other quality attributes.

2.12. Assumptions

List any assumptions that were made in the analysis that led to accepting this use case into the product description and writing the use case description.

2.13. Notes and Issues

List any additional comments about this use case or any remaining open issues or TBDs (To Be Determineds) that must be resolved. Identify who will resolve each issue, the due date, and what the resolution ultimately is.

Use Case Template

Use Case ID:			
Use Case Name:			
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	
Description:	
Preconditions:	
Postconditions:	
Priority:	
Frequency of Use:	
Flow of Events:	
Alternative Flows:	
Exceptions:	
Includes:	
Special Requirements:	
Assumptions:	
Notes and Issues:	

Use Case ID:	UC-1.1		
Use Case Name:	Search and Filter School Directory		
Created By:	Max	Last Updated By:	Paul
Date Created:	04 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	All Users (Initiating Actor)
Description:	Actors are able to select an educational tier, search the school directory and apply relevant filters (e.g., governance, courses offered, distance from a postal code) to narrow the search results.
Preconditions:	- The actor is on the school directory page.
Postconditions:	- A filtered and sorted paginated search result or an empty result if there are no matches.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user navigates to the school directory page. 2. The system displays the educational tier selector (Primary, Secondary, Post-Secondary/Tertiary, University). 3. The user selects an educational tier. 4. The system displays the search bar, the filter panel, the sort controls and a paginated list (10 schools per page) of summary cards for the tier. Each card shows the school name, school logo and location. 5. The user optionally enters a keyword or sets filters (governance type, course/programme for tertiary institutions, and distance from a postal code). 6. The system validates and sanitizes user input and queries the database. 7. The system displays a filtered view of the schools. 8. The user optionally changes the sort order (A–Z, Z–A, or distance ascending). 9. The system displays a sorted view of the results.
Alternative Flows:	- AF-S5a: If the distance filter is used with a postal code that is not a valid 6-digit string, an inline error “Please enter a valid 6-digit Singapore postal code.” will be displayed. Return to S5. - AF-S5b: If the postal code has the right format but cannot be located, an inline error “This postal code could not be located. Please check it and try again.” will be displayed. Return to S5. - AF-S6a: If the keyword has fewer than 2 characters, the system does not run the search, displays “Enter at least 2 characters to search.”, and the current results remain unchanged. Return to S5. - AF-S6b: If the keyword contains no letters or numbers (only spaces, symbols or code syntax such as “<script>“ or “;--”), the system does not run the search and displays “Search must include letters or numbers. Symbols and code syntax are ignored.” Return to S5. - AF-S6c: If the keyword contains disallowed characters in addition to letters or numbers, the system removes them, displays “Unsupported characters were removed. Letters,

	numbers, spaces, hyphens, periods and apostrophes only." and searches the remaining text. Resume with S7. - AF-S7: If there are no search results, the system will display "No schools match your search criteria. Try clearing filters or adjusting your search keywords." with a "Reset All Filters" button. Clicking the button restores all inputs to default and reloads the full school list for the tier. Return to S5.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error and return an empty table.
Includes:	None
Special Requirements:	- PERF-4: The system shall load and render the school directory within 2 seconds on visit or query. - QUAL-1: The school directory view shall be responsive and adjust to the screen size of the user. - SEC-4: The system shall sanitize the user input on the backend.
Assumptions:	1. The school database is already populated. 2. Distance filtering and sorting use the straight-line (Haversine) distance from the centre of the postal district given by the first two digits of the postal code. The system does not rely on a live external mapping API for distance calculations.
Notes and Issues:	None

Use Case ID:	UC-1.2		
Use Case Name:	View School Profile		
Created By:	Max	Last Updated By:	Paul
Date Created:	04 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	All Users (Initiating Actor)
Description:	Actors are able to view the school profile that contains additional information that is not visible from UC-1.1's search view.
Preconditions:	- The actor selects a school from the school directory page.
Postconditions:	- A populated profile page is displayed.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects a school from UC-1.1. 2. The system queries for the full school record. 3. The system displays the following information: ● School Identity (name, logo, governance type) ● Location (full address, 6-digit postal code, nearest MRT station with line code, nearest bus stops with stop name, 5-digit stop code and service numbers) ● Course & Curriculum (all offered tracks, diplomas, degrees or subject combinations) ● Itemized School Fees (per course, grouped by Singapore Citizen, Permanent Resident and International Student) ● Financial Aid Programs (e.g., MOE FAS, MOE Tuition Grant, Higher Education Bursary, fee waivers) ● Scholarship Details (internal and external, with eligibility requirements) ● Historical Statistics (multi-year entry data, e.g., 10th/90th percentile GPAs, JAE cut-off points, PSLE AL ranges, taken from the optional benchmarks file) ● Location Map (a mini map with a pin at the school's location; tiles loaded from OneMap through Leaflet, with a credit to the map provider) ● User Reviews (via UC-4.2)
Alternative Flows:	- AF-S3: If a section has no data, the system displays "Not available". Resume with S3. - AF-S3: If the school has no location data, the system displays "Not available" in place of the map. Resume with S3. - AF-S3: If the map fails to load (e.g., OneMap is unavailable), the map is not drawn and the rest of the profile is displayed normally. Resume with S3.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error.
Includes:	- UC-4.2
Special Requirements:	- BR-1: The system shall display course fee breakdowns based on citizenship status.

	- PERF-1: The system shall load and render the school profile within 2 seconds on visit or query. - QUAL-1: The school profile view shall be responsive and adjust to the screen size of the user.
Assumptions:	1. The school database is already populated. 2. Each school record holds its location data, which is used for the map pin. A school without location data shows "Not available" in place of the map (REQ-1.16). 3. Map tiles are requested by the user's browser directly from OneMap, not through the PathSG backend. If OneMap is down, only the map is affected. 4. Until a benchmarks file is loaded (UC-5.1), only the primary Phase 2B places taken and the 2017 secondary PSLE T-score cut-off are shown under Historical Statistics.
Notes and Issues:	None

Use Case ID:	UC-2.1		
Use Case Name:	Register User Account		
Created By:	Max	Last Updated By:	Paul
Date Created:	04 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Guest Users (Initiating Actor)
Description:	Actors are able to create a user account to save their preferences.
Preconditions:	- The actor is not authenticated. - The actor is on the registration page.
Postconditions:	- A user account is created and the actor is logged in; or - No user account is created and the form shows an inline error.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The guest user navigates to the registration page. 2. The system displays the registration form containing username and password fields. 3. The guest user enters a username and password. 4. The guest user clicks on the “Create Account” button. 5. The system sanitizes user input. 6. The system checks that the username is not already used. 7. The system checks that the password is sufficiently complex. 8. The system hashes the password and creates the account. 9. The system issues an access token and a refresh token, the frontend stores them in browser session storage to create the login session, and the system notifies the guest user that the account was created and they are logged in.
Alternative Flows:	- AF-S5a: If the user input contains special characters, an inline error "Username cannot contain special characters. Please use only letters and numbers." will be displayed on the form. Return to S3. - AF-S5b: If the username is longer than 36 characters, the server rejects it and an inline error “Username must be between 1 and 36 characters long and contain only letters and numbers.” will be displayed on the form. Return to S3. - AF-S6: If the username is already used (compared without regard to letter case), an inline error “This username is already taken. Please choose a different username.” will be displayed on the form. Return to S3. - AF-S7: If the password is not sufficiently complex, an inline error "Password must be at least 8 characters long and contain at least one number and one special character." will be displayed on the form. Return to S3.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error.
Includes:	None
Special Requirements:	- PERF-5: The system shall process the authentication request and return a response within 1s. - QUAL-2: The system shall show the inline errors within 0.3s. - SEC-1: The system shall not request, collect, or store any personally identifiable information (PII) in the database.

	- SEC-2: The system shall enforce password complexity (8 to 36 characters, 1 numeral, 1 special character) on the backend. - SEC-3: The system shall hash the password on the backend. - SEC-4: The system shall sanitize the user input on the backend. - SEC-5: The system shall handle user sessions using JSON Web Tokens (JWT). A short-lived access token (expires after 5 minutes) shall be sent in the Authorization: Bearer header of each protected request. A single-use refresh token (valid for 7 days, stored on the server only as a hash) shall be used to obtain a new token pair when the access token expires. The frontend shall keep the tokens in browser session storage, so the session ends when the browser tab is closed.
Assumptions:	1. The username is the unique identifier of a user account.
Notes and Issues:	Considerations 1. An optional email field for password recovery. 2. Additional password security measures such as salting.

Use Case ID:	UC-2.2		
Use Case Name:	Login User Account		
Created By:	Max	Last Updated By:	Paul
Date Created:	04 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Guest Users (Initiating Actor).
Description:	Actors are able to authenticate into their user account to access their saved preferences.
Preconditions:	- The actor is not authenticated. - The actor has an existing user account. - The actor is on the login page.
Postconditions:	- A user account is authenticated and the actor is logged in; or - The actor is not logged in and the form shows an inline error.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The guest user navigates to the login page. 2. The system displays the login form containing username and password fields. 3. The guest user enters a username and password. 4. The guest user clicks on the “Login” button. 5. The system sanitizes user input. 6. The system checks that the username and client IP address combination has made fewer than 5 login attempts in the last 5 minutes. 7. The system looks up the username, hashes the password and compares it with the stored hash. 8. The system issues an access token and a refresh token, the frontend stores them in browser session storage to create the login session, and the system notifies the guest user.
Alternative Flows:	- AF-S6: If the username and client IP address combination has already made 5 login attempts in the last 5 minutes, the system rejects the attempt (HTTP 429) and an inline error “Too many login attempts. Please wait a few minutes and try again.” will be displayed on the form. Return to S3 once the 5 minutes have passed. - AF-S7a: If the username does not exist, an inline error “No account exists with this username. Please check your username or create a new account.” will be displayed on the form. Return to S3. - AF-S7b: If the password does not match, an inline error “The password entered is incorrect. Please try again.” will be displayed on the form. Return to S3.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error.
Includes:	None
Special Requirements:	- PERF-5: The system shall process the authentication request and return a response within 1s. - QUAL-2: The system shall show the inline errors within 0.3s.

	- SEC-4: The system shall sanitize the user input on the backend. - SEC-5: The system shall handle user sessions using JSON Web Tokens (JWT). A short-lived access token (expires after 5 minutes) shall be sent in the Authorization: Bearer header of each protected request. A single-use refresh token (valid for 7 days, stored on the server only as a hash) shall be used to obtain a new token pair when the access token expires. The frontend shall keep the tokens in browser session storage, so the session ends when the browser tab is closed. - SEC-7: The system shall limit repeated login requests to 5 per username and client IP address per 5 minutes to reduce password guessing. Each allowed attempt counts, whether it succeeds or fails; attempts rejected by the limit do not count.
Assumptions:	1. The username is the unique identifier of a user account.
Notes and Issues:	None

Use Case ID:	UC-2.3		
Use Case Name:	Logout User Account		
Created By:	Max	Last Updated By:	Paul
Date Created:	04 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to logout from their user account and return to a guest user.
Preconditions:	- The actor is authenticated.
Postconditions:	- The session is terminated and the actor returns to guest user.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The authenticated user clicks on the “Logout” button. 2. The system invalidates the session by revoking the stored refresh token. 3. The system clears the access token and refresh token from browser session storage. 4. The system redirects the guest user to root.
Alternative Flows:	- AF-S2: The refresh token cannot be revoked on the server. The system proceeds to clear the tokens from browser session storage. Resume with S3.
Exceptions:	None
Includes:	None
Special Requirements:	SEC-5: The system shall handle user sessions using JSON Web Tokens (JWT). A short-lived access token (expires after 5 minutes) shall be sent in the Authorization: Bearer header of each protected request. A single-use refresh token (valid for 7 days, stored on the server only as a hash) shall be used to obtain a new token pair when the access token expires. The frontend shall keep the tokens in browser session storage, so the session ends when the browser tab is closed.
Assumptions:	1. Clearing the tokens from browser session storage ends the session on the client. The session also ends when the browser tab is closed.
Notes and Issues:	None

Use Case ID:	UC-2.4		
Use Case Name:	View User Profile		
Created By:	Max	Last Updated By:	Paul
Date Created:	05 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to configure their profile (target education level, academic score and residential postal code) to personalize their experience on the system. The saved postal code becomes the default origin point for all distance calculations.
Preconditions:	- The actor is authenticated. - The actor is on the profile page.
Postconditions:	- The profile is saved and the postal code is set as the default origin for proximity calculations; or - The profile is not saved and the form shows an inline error.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The authenticated user navigates to the profile page. 2. The system displays the target education level selector. 3. The authenticated user selects a target education level. 4. The system displays only the relevant fields for the education level: ● Primary: “None” or “Kindergarten completion” ● Secondary: PSLE AL score ● Post-Secondary: GCE O-Level L1R5 or ELR2B2 score ● University: Polytechnic CGPA or GCE A-Level Rank Points (RP) 5. The authenticated user enters their residential postal code (Update Postal Code) and fills in the relevant fields for the education level (Update Academic Result). 6. The authenticated user clicks on the “Save Profile” button. 7. The system sanitizes user input. 8. The system validates the postal code as a 6-digit numeric string and checks that it can be located. 9. The system validates the academic score against the range for the selected education level. 10. The system saves the profile, sets the postal code as the default reference point for proximity calculations, and notifies the authenticated user.
Alternative Flows:	- AF-S8a: If the postal code is not valid, an inline error “Please enter a valid 6-digit Singapore postal code.” will be displayed on the form. Return to S5. - AF-S8b: If the postal code has the right format but cannot be located, an inline error “This postal code could not be located. Please check it and try again.” will be displayed on the form. Return to S5. - AF-S9: If the academic score is not valid, an inline error beginning “Invalid academic score!” followed by the rule

	broken (e.g., “PSLE AL must be a whole number from 4 to 32.”, “GPA must be between 0.00 and 4.00.”, “Rank points must be between 0 and 90.”) will be displayed on the form. Return to S5.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error. - EX-2: The actor is not authenticated, or the access token has expired and the refresh token is rejected. The system logs the actor out and redirects them to the login page (UC-2.2). The use case ends.
Includes:	None
Special Requirements:	- BR-3: The system shall validate academic score boundaries (PSLE AL: whole number 4-32; O-Level L1R5: whole number 2-54; O-Level ELR2B2: whole number of 0 or more; Polytechnic GPA: 0.00-4.00; A-Level RP: 0-90). For the Post-Secondary and University tracks, at least one of the listed scores must be entered. - BR-4: The system shall validate postal code user input as a valid 6-digit Singapore postal code. - PERF-2: The system shall render the education fields within 1s of choosing an education level. - QUAL-2: The system shall show the inline errors within 0.3s. - SEC-1: The system shall not request, collect, or store any personally identifiable information (PII) in the database. - SEC-4: The system shall sanitize the user input on the backend.
Assumptions:	1. The authenticated user is truthful with their saved preferences. 2. Saving the postal code does not rely on a live external mapping API. Distances use the centre of the postal district (see UC-1.1).
Notes and Issues:	None

Use Case ID:	UC-2.5		
Use Case Name:	Bookmark School		
Created By:	Max	Last Updated By:	Paul
Date Created:	05 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to add a school bookmark so it persists in their saved list across sessions.
Preconditions:	- The actor is authenticated. - The actor is on a school summary card or school profile page.
Postconditions:	- The school is added into the saved list; or - The school is not added and the system shows a popup error.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The authenticated user clicks on the bookmark icon. 2. The system adds the school into the saved list. 3. The system changes the bookmark icon to show new state.
Alternative Flows:	None
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error. - EX-2: The actor is not authenticated, or the access token has expired and the refresh token is rejected. The system logs the actor out and redirects them to the login page (UC-2.2). The use case ends.
Includes:	None
Special Requirements:	- BR-2: The system shall allow bookmark management function only to authenticated users.
Assumptions:	1. There is no limit to the number of bookmarks. 2. The bookmark icon is not shown to guest users.
Notes and Issues:	None

Use Case ID:	UC-2.6		
Use Case Name:	Unbookmark School		
Created By:	Max	Last Updated By:	Paul
Date Created:	05 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to remove a school bookmark so it no longer persists in their saved list across sessions.
Preconditions:	- The actor is authenticated. - The school is bookmarked. - The actor is on the saved list page, a school summary card or a school profile page.
Postconditions:	- The school is removed from the saved list; or - The school is not removed and the system shows a popup error.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The authenticated user clicks on the “Remove” control on the school’s card in the saved list, or on the bookmark icon on a school summary card or school profile page. 2. The system removes the school from the saved list. 3. The system updates the view in real time: the card disappears from the saved list, or the bookmark icon changes to show the new state.
Alternative Flows:	None
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error. - EX-2: The actor is not authenticated, or the access token has expired and the refresh token is rejected. The system logs the actor out and redirects them to the login page (UC-2.2). The use case ends.
Includes:	None
Special Requirements:	- BR-2: The system shall allow bookmark management function only to authenticated users.
Assumptions:	1. There is no confirmation dialog for removing a bookmark. 2. There is no undo option after removing a bookmark.
Notes and Issues:	None

Use Case ID:	UC-2.7		
Use Case Name:	View Bookmarked Schools		
Created By:	Max	Last Updated By:	Paul
Date Created:	05 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to view all their bookmarked schools in one place.
Preconditions:	- The actor is authenticated. - The actor is on the saved list page.
Postconditions:	- The saved list is displayed; or - An empty state is displayed.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The authenticated user navigates to the saved list page. 2. The system retrieves all the schools from the saved list. 3. The system displays a quick-view card for each school: ● School Name and School Logo ● Governance Type (e.g., Government, Independent) ● Distance in km (the straight-line distance from the user's saved postal code to the school, measured from the centre of the postal district) ● Primary Entry Benchmark for the school's tier (e.g., PSLE AL range, JAE cut-off point, or Polytechnic IGP GPA)
Alternative Flows:	- AF-S2: If the saved list is empty, the system displays an empty state with a link back to the school directory. - AF-S3: If there are no postal code saved, the system omits the distance. Resume with S3.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error. - EX-2: The actor is not authenticated, or the access token has expired and the refresh token is rejected. The system logs the actor out and redirects them to the login page (UC-2.2). The use case ends.
Includes:	None
Special Requirements:	- BR-2: The system shall allow bookmark management function only to authenticated users. - PERF-3: The system shall compute the distance for all schools in the saved list within 2s. - QUAL-1: The saved list view shall be responsive and adjust to the screen size of the user.
Assumptions:	1. The saved list is of a realistic number to satisfy PERF-3. 2. The saved list has no map. The SRS only places maps on the school profile and roadmap pages.
Notes and Issues:	None

Use Case ID:	UC-3.1		
Use Case Name:	Generate Education Roadmap		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to generate an education roadmap using both the rule-based method (ranked by grade or by distance) and the AI-generated method (OpenAI API). Each plan is shown with its own map (Leaflet with OneMap tiles) of the user's postal code and the recommended schools. The actor can save the preferred plan in their user account.
Preconditions:	- The actor is authenticated. - The actor is on the roadmap page.
Postconditions:	- The preferred plan is saved as the roadmap and a confirmation notice is displayed; or - No plan is saved and the system shows an error.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The authenticated user navigates to the roadmap page. 2. The system queries for the saved preferences of the authenticated user (postal code, current education level and current education result). 3. The system displays the target transition stage selector with the transition stage preselected using the saved preferences. 4. The system displays the Rule-Based Plan controls. For stages that offer both options (Primary → Secondary, Secondary → Post-Secondary) this is a "Rank by" choice (Distance or Grade, Grade by default) and, when Grade is selected, an "Acceptable distance" slider (1 km to 20 km, in 1 km steps). The panel also has an "About" button. 5. The authenticated user optionally changes the target transition stage, the "Rank by" choice or the distance slider. 6. The authenticated user clicks on the "Generate Roadmap" button. 7. The system verifies that a valid 6-digit postal code is stored in the profile. 8. The system executes the rule set for the selected transition stage, following the chosen ranking priority, and displays the rule-based plan while showing a loading state in the AI panel. 9. The system invokes the module to create the AI-generated plan asynchronously and displays it when complete. The prompt contains the student's stage, score, postal code and a filtered list of candidate schools, and the username is never sent. The backend keeps only valid school IDs from the response. 10. The system displays a map in each plan panel, showing a pin for the user's postal code and a numbered pin for each recommended school (1st, 2nd, 3rd choice, matching the plan

	list order). Tiles are loaded from OneMap and a credit to the map provider is shown. 11. The authenticated user clicks on the "Save Roadmap" button of their preferred plan. 12. The system stores the preferred plan in the database. 13. The system displays a notice banner confirming the action (e.g., "Roadmap saved to your profile.").
Alternative Flows:	- AF-S4: If the stage is P1 Registration, the "Rank by" choice is not shown and the plan is ranked by distance only. If the stage is No School → Preschool or Post-Secondary → University, no rule-based plan is generated (the AI-generated plan still is). Resume with S5. - AF-S5: If the authenticated user changes the target transition stage, the system updates the Rule-Based Plan controls for the selected stage. Resume with S4. - AF-S7: If no postal code is found, the system creates a popup "Missing Profile Information: Please enter your postal code in your profile before generating a roadmap." with a link to the profile page (UC-2.4). The use case ends. - AF-S8a: If no school is within the acceptable distance, the Rule-Based Plan panel displays "No schools are within X km of [postal code]. Try a larger distance." and the "Save Roadmap" button is disabled for that plan. Resume with S9. - AF-S8b: If the rule-based plan cannot be generated, the Rule-Based Plan panel displays "Unable to generate the rule-based plan at this time. Please try again." with a Retry button. The AI-generated plan is not affected. Resume with S9. - AF-S9a: If the AI module errors or exceeds 15s, the system displays "Unable to generate AI recommendation at this time. Please try again." with a "Retry AI Generation" button in the AI panel only. The same message is shown for every AI failure, including timeouts and rate-limit responses. Retry re-runs the AI call asynchronously without regenerating the rule-based plan or reloading the page. Resume with S9. - AF-S9b: If the actor has made more than 10 AI generation requests in 5 minutes (each "Generate Roadmap" and "Retry AI Generation" counts), the system rejects the AI request (HTTP 429) and the AI panel displays the standard message "Unable to generate AI recommendation at this time. Please try again." with the "Retry AI Generation" button. The rule-based plan is unaffected. Resume with S11. - AF-S10a: If a recommended school has no location data, it is not pinned and is listed under the map as "location not available". Resume with S11. - AF-S10b: If the map fails to load (e.g., OneMap is unavailable), the plan list is still displayed and unaffected. Resume with S11. - AF-S10c: If the actor changes the "Rank by" choice or releases the distance slider, the system re-ranks only the Rule-Based Plan and its map without reloading the page or requesting the AI plan again. Resume with S11.

	- AF-S10d: If the actor clicks “About”, the system opens a dialog explaining how the plan is ranked for the chosen stage. The dialog closes with its button or the Esc key. Resume with S11. - AF-S12: If there is an existing roadmap, the system displays a confirmation dialog warning that it will be overwritten. If the actor confirms, resume with S12. If the actor cancels, the system closes the dialog without saving. Return to S11.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error.
Includes:	None
Special Requirements:	- PERF-6: The system shall display the rule-based plan within 2s of generation. - PERF-7: The system shall display the AI-generated plan within 15s of generation. - QUAL-1: The roadmap view shall be responsive and adjust to the screen size of the user. - QUAL-2: The system shall show the inline errors within 0.3s. - QUAL-6: The system shall render the rule-based plan without waiting for the AI-generated plan. - QUAL-7: The system shall display module errors within the panel without reloading the page. - SEC-7: The system shall limit AI roadmap generation to 10 requests per account per 5 minutes. Rule-based generation is not limited.
Assumptions:	1. A dedicated rule set exists for each stage that has a Rule-Based Plan (P1 Registration, Primary → Secondary, Secondary → Post-Secondary). 2. Each user account holds only one saved roadmap at any time. 3. The OpenAI API and OneMap are available. If either is down, only the AI plan or the maps are affected. 4. Distances are computed locally (Haversine, from the centre of the postal district), not through OneMap.
Notes and Issues:	None

Use Case ID:	UC-3.2		
Use Case Name:	View Saved Roadmap		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to view the roadmap saved in their user account.
Preconditions:	- The actor is authenticated. - The actor is on the roadmap page.
Postconditions:	- The saved roadmap is displayed; or - An empty state is displayed.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The authenticated user navigates to the roadmap page. 2. The system queries for the roadmap of the authenticated user. 3. The system displays the roadmap containing the following: ○ Target Transition Stage ○ School Choices 4. The system displays a map of the saved roadmap, with a pin for the user's postal code and a numbered pin for each school choice (same pins as the generated plan). 5. The system displays the "Generate Roadmap", "Edit Roadmap" and "Delete Roadmap" buttons.
Alternative Flows:	- AF-S2: If no roadmap is saved, the system displays an empty state with a "Generate Roadmap" button. The use case ends. - AF-S4: If a school choice has no location data, it is not pinned and is listed under the map as "location not available". Resume with S5. - AF-S4: If the map fails to load, the roadmap details and buttons are still displayed. Resume with S5.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error. - EX-2: The actor is not authenticated, or the access token has expired and the refresh token is rejected. The system logs the actor out and redirects them to the login page (UC-2.2). The use case ends.
Includes:	None
Special Requirements:	- QUAL-1: The roadmap view shall be responsive and adjust to the screen size of the user.
Assumptions:	1. Each user account holds only one saved roadmap at any time.
Notes and Issues:	None

Use Case ID:	UC-3.3		
Use Case Name:	Edit Saved Roadmap		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to edit the roadmap saved in their user account.
Preconditions:	- The actor is authenticated. - The actor has a saved roadmap. - The actor is on the roadmap page.
Postconditions:	- The saved roadmap is updated; or - The roadmap update fails and the system shows a popup error.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The authenticated user clicks on the “Edit Roadmap” button of the saved roadmap. 2. The system displays a dropdown for each school choice, populated only from valid institutional options. 3. The authenticated user modifies the school choices. 4. The authenticated user clicks on the “Save Changes” button. 5. The system sanitizes user input and checks the school choices against the schools allowed for the stage. 6. The system updates the saved roadmap and displays the updated roadmap, with the map pins updated to the new school choices.
Alternative Flows:	- AF-S3: If the authenticated user clicks on the “Cancel” button, the system discards the changes and restores the original saved roadmap view. The use case ends.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error. - EX-2: The actor is not authenticated, or the access token has expired and the refresh token is rejected. The system logs the actor out and redirects them to the login page (UC-2.2). The use case ends.
Includes:	None
Special Requirements:	- SEC-4: The system shall sanitize the user input on the backend.
Assumptions:	1. The system does not verify if the school is a suitable fit. 2. The dropdown is populated only from the schools that are valid for the stage.
Notes and Issues:	None

Use Case ID:	UC-3.4		
Use Case Name:	Delete Saved Roadmap		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to delete the roadmap saved in their user account.
Preconditions:	- The actor is authenticated. - The actor has a saved roadmap. - The actor is on the roadmap page.
Postconditions:	- The roadmap is deleted from the user account; or - The roadmap deletion fails and the system shows a popup error.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The authenticated user clicks on the “Delete Roadmap” button of the saved roadmap. 2. The system displays a confirmation dialog. 3. The authenticated user confirms the deletion. 4. The system removes the roadmap from the database. 5. The system displays the empty state of the roadmap page.
Alternative Flows:	- AF-S3: If the authenticated user cancels the confirmation dialog, the system closes the dialog. The use case ends.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup to notify the user of the error. - EX-2: The actor is not authenticated, or the access token has expired and the refresh token is rejected. The system logs the actor out and redirects them to the login page (UC-2.2). The use case ends.
Includes:	None
Special Requirements:	None
Assumptions:	1. There is no undo option after deleting a roadmap.
Notes and Issues:	None

Use Case ID:	UC-4.1		
Use Case Name:	Submit School Review		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to submit a review on a school profile page.
Preconditions:	- The actor is authenticated. - The actor is on a school profile page.
Postconditions:	- The review is created and visible from the school profile; or - The review creation fails and the system shows an inline error.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The authenticated user navigates to a school profile page. 2. The system checks for a review by the authenticated user. 3. The system displays a review text area with a live word counter (e.g., “40 / 200 words”). 4. The authenticated user types a review into the review text area. 5. The authenticated user clicks on the “Submit Review” button. 6. The system removes HTML tags from the review and trims leading and trailing spaces. 7. The system checks that the review is not empty and does not exceed 200 words. 8. The system saves the review and refreshes the view.
Alternative Flows:	- AF-S2: If a school has already been reviewed, the system displays their existing review with the “Edit Review” and “Delete Review” buttons instead. The use case ends. - AF-S4: If the review reaches 200 words, the text area keeps only the first 200 words and drops any words beyond the limit. Resume with S5. - AF-S5: If the actor is a guest user who attempts to submit a review, the system blocks the submission (HTTP 401) and displays a warning modal “Please log in or create an account to post a review.” The use case ends. - AF-S7a: If the review is blank or consists entirely of spaces, an inline error "Review text cannot be blank or consist entirely of spaces." will be displayed on the form. Return to S4. - AF-S7b: If the review exceeds 200 words, the system rejects it and an inline error “Review text must not exceed 200 words.” will be displayed on the form. Return to S4. - AF-S8: If the actor already has a review for the school, the system rejects the submission and displays “You have already reviewed this school.” Return to S2.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup “Unable to complete your request at this time. Please try again later.” - EX-2: The access token has expired and the refresh token is rejected. The system logs the actor out and asks them to log in again.
Includes:	None

Special Requirements:	- BR-6: The system shall allow only one review per school from a single user account. - QUAL-2: The system shall show the inline errors within 0.3s. - SEC-4: The system shall sanitize the user input on the backend.
Assumptions:	1. The reviews are displayed immediately without moderation. 2. The review form is visible to guest users, but submission requires authentication.
Notes and Issues:	Considerations 1. A moderation flow for user review.

Use Case ID:	UC-4.2		
Use Case Name:	View School Reviews		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	All Users (Initiating Actor).
Description:	Actors are able to view all reviews on a school profile page.
Preconditions:	- The actor is on a school profile page.
Postconditions:	- The reviews are displayed; or - An empty state is displayed.
Priority:	Medium
Frequency of Use:	High
Flow of Events:	1. The user navigates to a school profile page. 2. The system retrieves all the reviews for a school. 3. The system displays the following information for each review: o Reviewer Username o Submission Time o Review Content
Alternative Flows:	- AF-S2: If there are no reviews of the school, the system displays an empty state "No reviews yet". The use case ends.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup “Unable to complete your request at this time. Please try again later.”
Includes:	None
Special Requirements:	- PERF-1: The system shall load and render the school reviews within 2 seconds on visit or query. - QUAL-1: The review section shall be responsive and adjust to the screen size of the user.
Assumptions:	1. Reviews are sorted from newest to oldest. 2. Reviews are shown to guest users.
Notes and Issues:	Considerations 1. Pagination if there are too many reviews.

Use Case ID:	UC-4.3		
Use Case Name:	Edit School Review		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to edit their review of a school profile page.
Preconditions:	- The actor is authenticated. - The actor has reviewed the school. - The actor is on a school profile page.
Postconditions:	- The review is updated and visible from the school profile; or - The review update fails and the system shows an inline error.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. The authenticated user clicks on the “Edit Review” button of their own review. 2. The system displays the review text area populated with the existing review and a word counter. 3. The authenticated user modifies the review. 4. The authenticated user clicks on the “Update Review” button. 5. The system sanitizes user input. 6. The system checks that the review is not empty. 7. The system updates the review and refreshes the view.
Alternative Flows:	- AF-S3: If the review reaches 200 words, the text area keeps only the first 200 words and drops any words beyond the limit. Resume with S4. - AF-S4: If the authenticated user clicks on the “Cancel” button, the system discards all modified changes and restores the original review text display. The use case ends. - AF-S6a: If the review is blank or consists entirely of spaces, an inline error "Review text cannot be blank or consist entirely of spaces." will be displayed on the form. Return to S3. - AF-S6b: If the review exceeds 200 words, the system rejects it and an inline error “Review text must not exceed 200 words.” will be displayed on the form. Return to S3.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup “Unable to complete your request at this time. Please try again later.” - EX-2: The access token has expired and the refresh token is rejected. The system logs the actor out and asks them to log in again.
Includes:	None
Special Requirements:	- QUAL-2: The system shall show the inline errors within 0.3s. - SEC-4: The system shall sanitize the user input on the backend.
Assumptions:	1. The edit button is shown only on the review created by the authenticated user.
Notes and Issues:	None

Use Case ID:	UC-4.4		
Use Case Name:	Delete School Review		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	05 Nov 2026

Actor:	Authenticated Users (Initiating Actor).
Description:	Actors are able to delete their review of a school profile page.
Preconditions:	- The actor is authenticated. - The actor has reviewed the school. - The actor is on a school profile page.
Postconditions:	- The review is deleted from the school profile page; or - The review deletion fails and the system shows a popup error.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. The authenticated user clicks on the “Delete Review” button of their own review. 2. The system displays a confirmation dialog. 3. The authenticated user confirms the deletion. 4. The system removes the review from the database. 5. The system displays the review text area.
Alternative Flows:	- AF-S3: If the authenticated user cancels the confirmation dialog, the system closes the dialog. The use case ends.
Exceptions:	- EX-1: The database is unreachable. The system will create a popup “Unable to complete your request at this time. Please try again later.” - EX-2: The access token has expired and the refresh token is rejected. The system logs the actor out and asks them to log in again.
Includes:	None
Special Requirements:	None
Assumptions:	1. The delete button is shown only on the review created by the authenticated user.
Notes and Issues:	None

Use Case ID:	UC-5.1		
Use Case Name:	Update School Database		
Created By:	Max	Last Updated By:	Paul
Date Created:	12 Sep 2026	Date Last Updated:	08 Nov 2026

Actor:	Administrators (Initiating Actor).
Description:	Actors are able to manually update the school database by uploading a CSV file containing the latest school dataset obtained from external sources such as “data.gov.sg”.
Preconditions:	- The actor is authenticated as an administrator. - The actor is on the admin panel.
Postconditions:	- The school database is updated with the latest dataset; or - The school database is unchanged and the system shows a popup error.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The administrator navigates to the admin panel. 2. The system displays a file upload control, an “Update Dataset” button, and the update status (date of last successful update, number of records, and whether an update is running: Idle or Update running). 3. The administrator selects the latest school dataset as a CSV file, previously downloaded from an external source such as data.gov.sg. 4. The system validates the file (type, non-empty, and 20 MB or less) and displays its name. 5. The administrator clicks on the “Update Dataset” button. 6. The system disables further interactions with the file upload control and “Update Dataset” button, and the button reads “Updating dataset…”. 7. The system recognises the dataset kind from the file’s column headers, parses the CSV records and standardizes them to the database schema (trimming text, handling empty values, converting numeric fields, splitting list fields). 8. The system creates a single transaction to bulk write the records, adding new schools and updating existing schools matched by their unique school identifier. 9. The system commits the transaction, re-enables the control and button, and displays “Database updated successfully with the latest Singapore school data”.
Alternative Flows:	- AF-S3: If no CSV file is selected, the system does not start the update and displays an inline error “Please select a CSV file before updating the database.” Return to S3. - AF-S4a: If the selected file is not a .csv file, is empty, or exceeds 20 MB, the system rejects it without processing and displays an inline error “Invalid file. Please upload a non-empty .csv file within the size limit.” Return to S3. - AF-S4b: If the administrator selects a different file before clicking “Update Dataset”, the system validates it and displays the new file name. Return to S4.

	- AF-S5: If the administrator clicks “Update Dataset” while another update is still running, the system does not start a second update and shows beside the form “A database update is already running. Please wait for it to finish.” The use case ends. - AF-S7: If an error occurs during parsing or standardization, the system aborts the update, rolls back any pending changes, re-enables the controls, and displays “Database Update Failed: Unable to process dataset format. Please check source data and try again.” Return to S3.
Exceptions:	- EX-1: The database is unreachable. The system aborts the update, re-enables the controls, and creates a popup “Database Update Failed: Unable to connect to local database. Please try again later.”
Includes:	None
Special Requirements:	- BR-5: The system shall allow database update functions only to administrator accounts. - PERF-8: The system shall complete the operation within 10s.
Assumptions:	1. The administrator obtains a CSV file that follows the expected dataset schema. 2. The administrator role is assigned directly in the database. 3. The system does not fetch data from external endpoints. The administrator downloads the CSV beforehand. 4. Each school record in the CSV carries location data for map pins and distances. Records without it are treated as having no location data. 5. A dataset kind that is not uploaded keeps its last upload, or the bundled copy when none exists.
Notes and Issues:	None

Use Case ID:	UC-5.2		
Use Case Name:	Export School Database		
Created By:	Max	Last Updated By:	Paul
Date Created:	26 Oct 2026	Date Last Updated:	08 Nov 2026

Actor:	Administrators (Initiating Actor).
Description:	Actors are able to export the entire school database, including all school records and associated details, into a CSV file. User accounts, saved roadmaps and reviews are excluded.
Preconditions:	- The actor is authenticated as an administrator. - The actor is on the admin panel.
Postconditions:	- A CSV file containing the current school database (no user or review data); or - No file is downloaded and the system shows a popup error.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The administrator navigates to the admin panel. 2. The administrator clicks on the “Export Database” button. 3. The system queries the database to retrieve all school records and associated details, excluding user account data, saved roadmaps and reviews. 4. The system formats the retrieved records into a CSV file named “pathsg_schools_YYYYMMDD.csv” using the date of export in Singapore time (UTC+8). 5. The system initiates the download of the CSV file to the administrator’s local machine through the browser.
Alternative Flows:	- AF-S3: If the school database is empty, the system aborts the export and displays a popup “Export Failed: Unable to retrieve school data at this time.” The use case ends.
Exceptions:	- EX-1: The database is unreachable. The system aborts the export and displays a popup “Export Failed: Unable to retrieve school data at this time.”
Includes:	None
Special Requirements:	- PERF-9: The system shall complete the export operation within 5s. - SEC-6: The system shall restrict database export function to authenticated administrator accounts.
Assumptions:	1. The administrator role is assigned directly in the database.
Notes and Issues:	None

Appendix A: Use Case Diagram

Use Case Diagram.png

If the above can't be opened, the file can be found within the supporting document folder.